A3: Blockwise Layout Design of an Integer-N Phase Locked Loop

EN4431: Analog IC Design

May 8, 2026

Group Project

The US CHIPS Act has allocated US$53 billion to this industry, most of which has been earmarked for chip fabrication [1]. The question is, who will design the chips that must fill the capacity of these fabrication lines? There is a shortage of chip design talent in the USA and around the globe who have practical experience in the entire chip design flow [2]. Gaining this experience takes years of training in a chip design lab or industry setting. According to Peter Kinget (Professor of Electrical Engineering, Columbia University, USA), "providing practical experience is crucial in order to teach IC design successfully" [3].

Figure 1 illustrates the development stages that the industry follows for chips of moderate complexity. Beginning with product definition and performance specifications, engineers embark upon the design of the chip and, in about six to nine months, complete the first "draft". The result is in the form of tens or hundreds of "schematics". Next, the chip goes to "layout"; i.e., designers create geometries (mostly squares and rectangles) that must be built on the actual chip. Since layout introduces its own "parasitics," the result may not meet the performance targets, requiring iteration with the design. Anticipating that the package in which the chip will be housed will contribute additional parasitics, we account for them in this iteration phase as well. This cycle takes three to six months. After the chip layout is completed and is verified to match the schematics, it is sent for fabrication, e.g., to Taiwan Semiconductor Manufacturing Co. (TSMC). Fabrication requires three to four months. Upon return, the chip is mounted in a suitable package and undergoes test and characterization; this phase takes another three to four months [1].

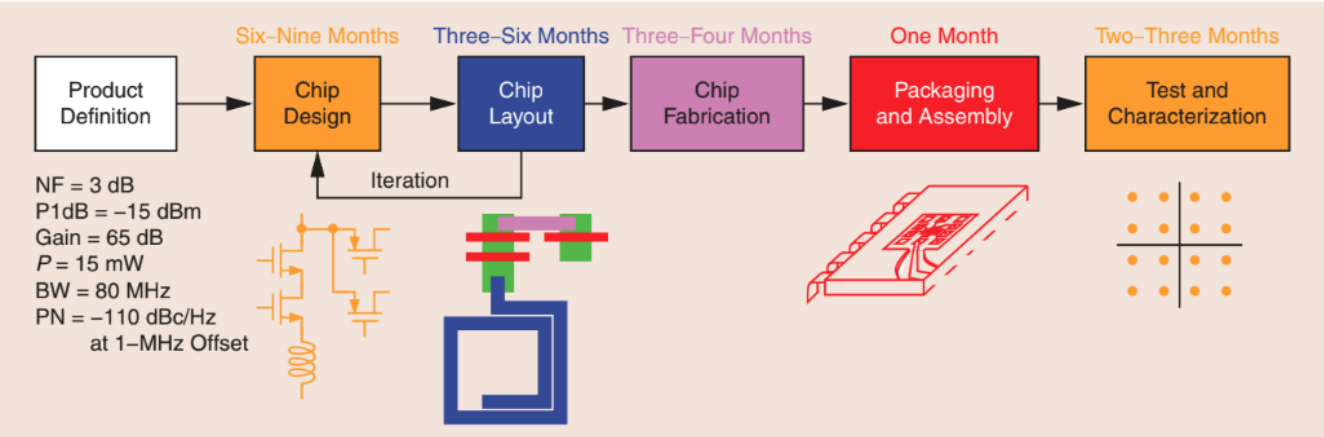


Figure 1: A typical chip development time frame [1].

Analog/RF/Mixed Signal IC Design Flow

We use an analog design flow as illustrated in Figure 2. As commercial design tools such as Virtuoso and Calibre now openly support the SKY130 open-source PDK, a fabricable layout can be designed. The following steps, as highlighted in Introduction to Cadence for Analog IC Design, can be followed for that:

- **Start Layout:** Create a pin schematic, symbol view, and testbench in preparation for layout.
- **Component Placement:** Identify components and place them at the appropriate location.
- **Routing:** Connecting components.
- **Design Rule Check:** Check if layout obeys design rules.
- **Layout vs. Schematic:** Check layout components are correctly connected.
- **Parasitic Extraction and Post-Layout Simulation:** Extracting parasitics with Assura and including it into simulation.

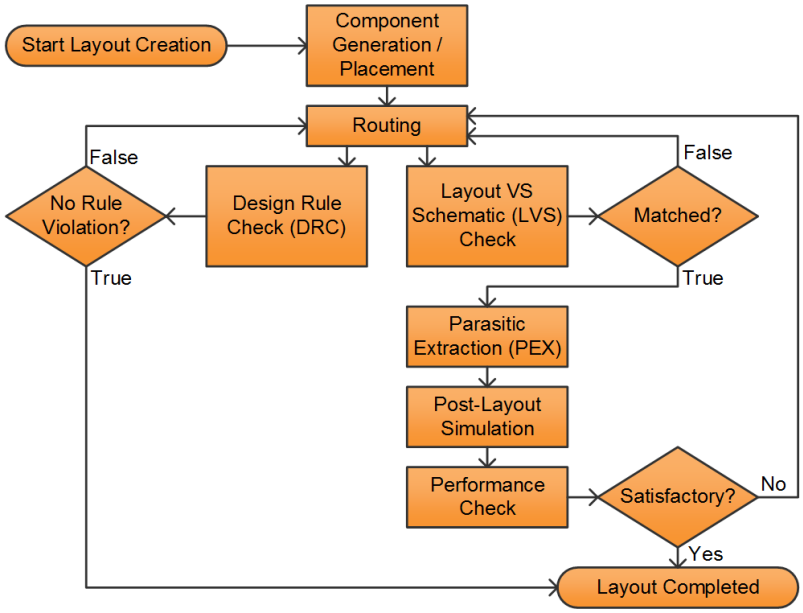


Figure 2: The typical analog layout design flow.

SKY130 PDK Layer Stack

The SkyWater open-source PDK is a collaboration between Google and SkyWater Technology Foundry to provide a fully open-source PDK and related resources for creating manufacturable designs at SkyWater's facility. This PDK is targeting the SKY130 process node. The following Figure 3 is the layer stack in the SKY130 process. The local interconnect, made of *TiN*, is quite characteristic of this process and is only used for very short-distance interconnects on devices.

The SKY130-Open-DesignLib documentation contains all the required information of the PDK. More information about process specifications and layout rules is also available in the same documentation.

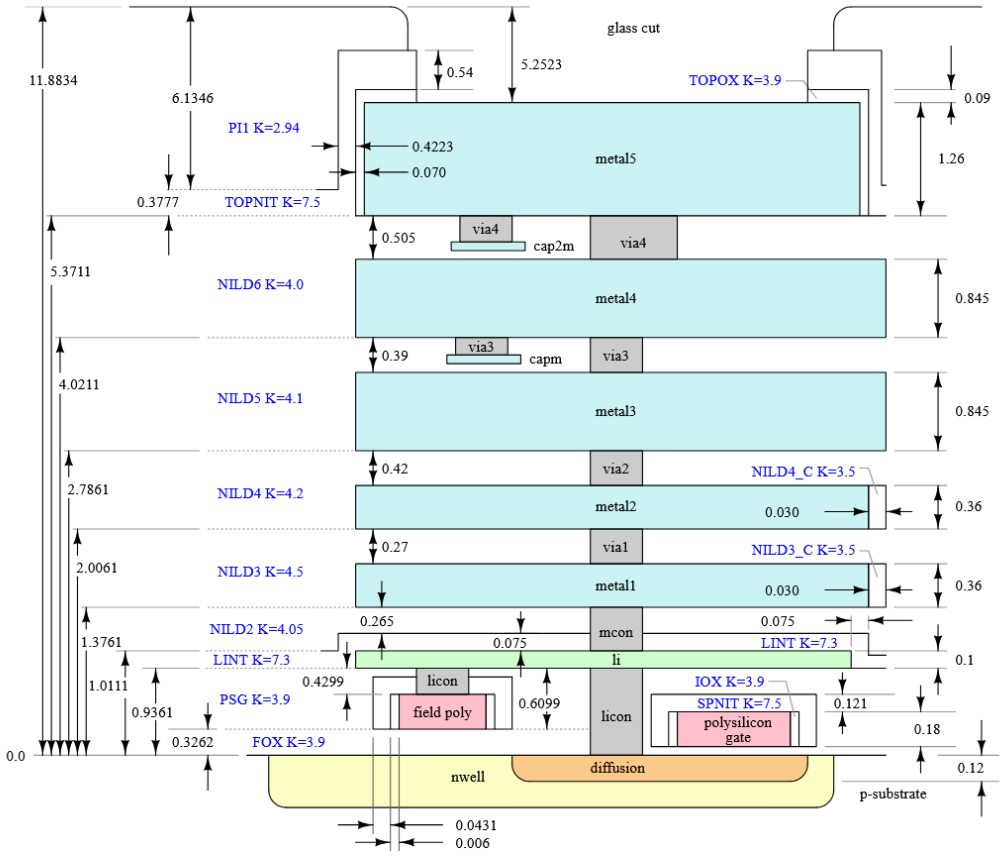


Figure 3: Cross-section layer stack of SKY130 process.

Task Description

Recall the circuit blocks you designed in your previous assignment (A2). In this assignment, you will be designing the layout for each of those blocks. For that, first you need to divide into groups of 2, where the work distribution is as follows:

- Group 01: Phase frequency detector, Charge pump, and Loop filter
- Group 02: Current-starved ring VCO
- Group 03: Programmable divider

The best designs among the group members can be used to design the layout. Alternatively, a combination of both can be used. You may also use the designs in the paper "A Ring-Oscillator-Based 2.4 GHz Integer-N PLL Design in SkyWater 130nm Open-Source Technology - ISPACS 2025" [4] as a reference. You are free to make any modifications you prefer to the design, such as updating the transistor sizes of the schematic. You can follow this Tutorial to familiarize yourself with the tool.

Once the projects have passed physical verification (DRC/LVS) and post-layout simulations after parasitic extraction (PEX), then the filler generation process is done. However, in this assignment, post-layout simulations after PEX and filler generation are not mandatory. After the final GDSII file has been generated and submitted, the selected designs will be processed at an advanced clean room facility. This process takes around 4 to 6 months, depending on the technology. After this process, the designers can obtain the chip die and package the die into QFN32 chips as shown in Figure 4.

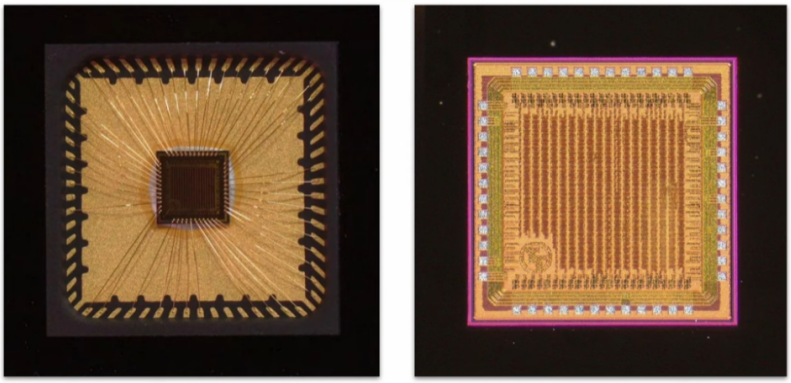


Figure 4: Wirebonded and packaged for measurement, obtained from TT.

The following figures 5 and 6 display the implementation details of the PLL project submitted to IHP TO_July tapeout run as a part of last year's EN4431 - Analog IC Design course.

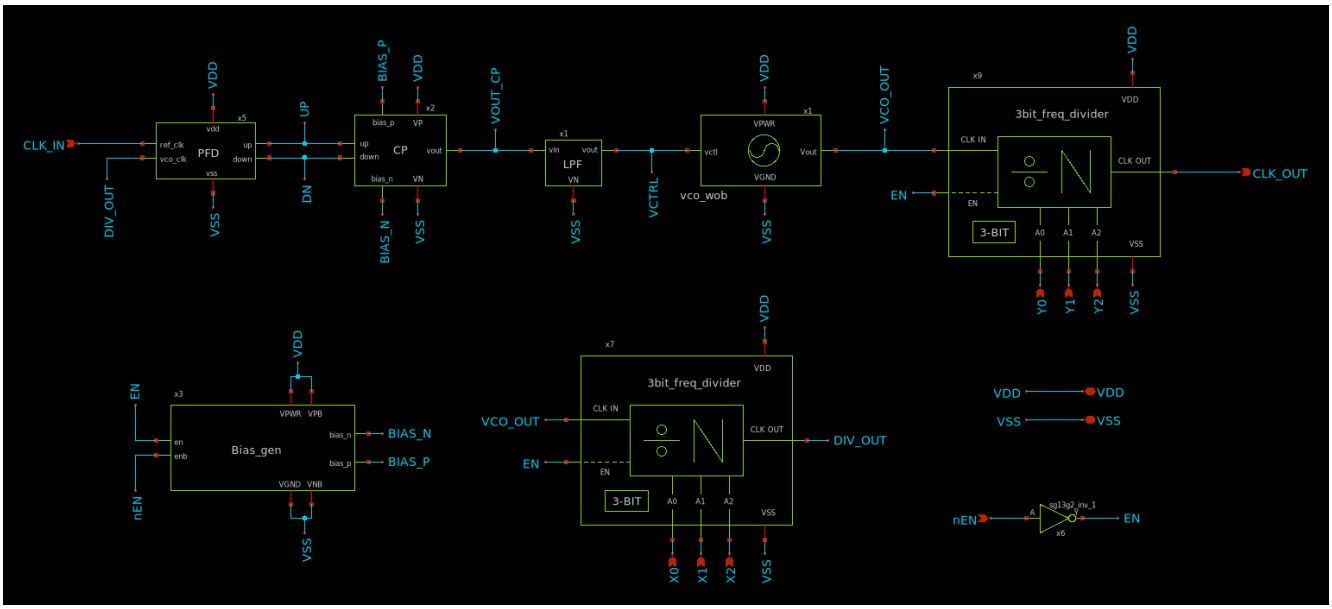


Figure 5: The top-level schematic design of 30 MHz Fractional-N PLL project from 2025.

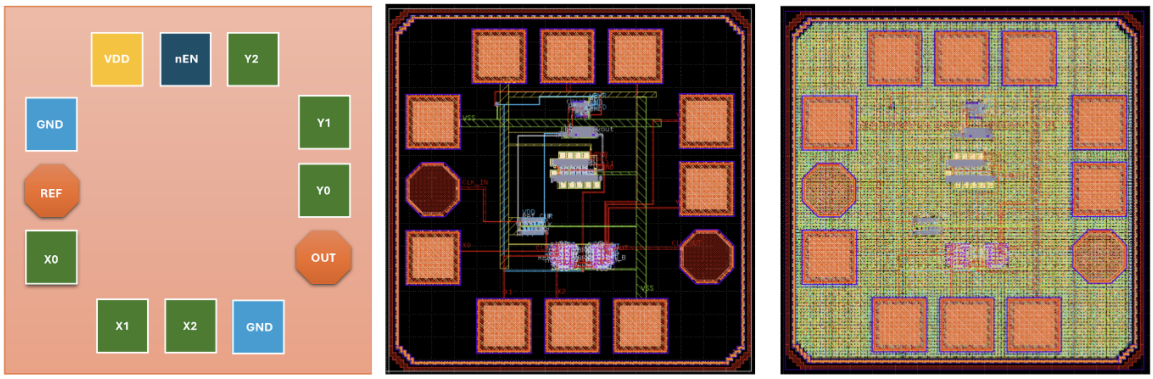


Figure 6: The pin map and the final layout from the 30 MHz Fractional-N PLL project from 2025.

Instructions

Reports must be PREPARED AND SUBMITTED INDIVIDUALLY. Each REPORT SHOULD CONTAIN ONLY THE INDIVIDUAL'S WORK on their chosen design. Details of the group design may be briefly mentioned in the final section. The design goals of this project are as follows:

- Organize your design files using the folder structure available in Submission template for Tiny Tapeout SKY130.
- The design should be as simple as possible to reduce the chance of failure.
- The design should be as small as possible so it can be incorporated into future Tiny Tapeout runs with minimal area overhead (less than 160 μm $\times$ 225 μm).

Due Date: 24th May 2026 by 11.59 PM
Target Tapeout Run: TTSKY26c (September 2026) [Link]

References

[1] B. Razavi, "Education of chip designers at a large scale: A proposal [society news]," *IEEE Solid-State Circuits Magazine*, vol. 16, no. 2, pp. 76–83, 2024.

[2] The Economist, "America is building chip factories. now to find the workers," *The Economist*. [Online]. Available: https://www.economist.com/united-states/2023/08/05/america-is-building-chip-factories-now-to-find-the-workers

[3] P. Kinget, "Teaching ic design: From concepts to testing a fabricated custom chip [society news]," *IEEE Solid-State Circuits Magazine*, vol. 15, no. 3, pp. 87–93, 2023.

[4] N. F. Assaify, A. N. Irfansyah, and T. Mujiono, "A ring-oscillator-based 2.4 ghz integer-n pll design in skywater 130nm open-source technology," in *2025 International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, 2025, pp. 1–5.